

SIMTiX: A Compact, Fully-Open SIMT Accelerator for RISC-V with an Energy-Aware Divergence Study

Krishna Gorai

Indian Institute of Technology Mandi

Email: kg05112001@gmail.com

Abstract—Open, readable hardware for studying Single-Instruction Multiple-Thread (SIMT) execution remains scarce: existing open GPGPU cores are large and difficult to bring up, while teaching cores rarely reach a real implementation flow. We present SIMTiX, a compact open-source SIMT accelerator attached to a five-stage RISC-V host through a memory-mapped offload interface. SIMTiX implements the full SIMT toolbox in a small, synthesizable design: eight lanes organized into hardware warps, a round-robin scheduler with a background memory engine for latency hiding, address coalescing on a cache-line-wide data port, a per-warp SIMT reconvergence stack for control divergence, and an on-chip scratchpad for data reuse. A key implementation result is that placing the vector register file and scratchpad in distributed RAM (LUTRAM) shrinks the design from 74% to 13% of the LUTs of a Xilinx Zynq UltraScale+ device while closing timing at 100 MHz. We characterize SIMTiX with a six-kernel benchmark suite spanning streaming, stencil, reduction, and data-dependent control-flow workloads, verified bit-exact against a golden model at sizes up to 1024 threads. Against the scalar host CPU executing the same work, SIMTiX achieves 7–10 $\times$ speedup on streaming kernels, 2.8 $\times$ on a divergence-heavy kernel, and—reported honestly—a slowdown on a serial reduction, exposing where the SIMT model does and does not pay off. Using the SIMT active mask as a per-lane clock-enable, we quantify the lane-datapath dynamic energy a divergence-aware gated design saves (up to 22.5% on a divergent kernel). The complete RTL, benchmark harness, and FPGA flow are publicly available.

Index Terms—SIMT, GPGPU, RISC-V, hardware accelerator, control divergence, memory coalescing, FPGA, clock gating, open-source hardware.

I. INTRODUCTION

Graphics processing units execute thousands of scalar threads in lockstep groups called *warps* under a Single-Instruction Multiple-Thread (SIMT) model. The microarchitectural mechanisms that make this efficient—warp scheduling for latency hiding, memory coalescing, control-divergence handling, and on-chip scratchpads—are well documented at the level of commercial GPUs, but they are hard to *study in RTL*. Production GPUs are proprietary; open GPGPU cores such as Vortex [1] are powerful but large and complex to bring up; and the small academic cores are often simulator-only or stop short of a real implementation flow.

This paper presents **SIMTiX**, a deliberately compact open-source SIMT accelerator for RISC-V whose entire path—RTL, simulation, synthesis, place-and-route, bitstream, and post-route gate-level sign-off—is reproducible on a single

workstation with free or widely available tools. SIMTiX is small enough to read end-to-end yet implements the complete SIMT mechanism set, and we use it to produce a measured, honest evaluation rather than a demonstration.

Contributions.

- A compact, fully-open SIMT accelerator (eight lanes, four resident warps) integrated with a five-stage RISC-V host as a self-contained system-on-chip, exercising the entire RTL-to-bitstream flow (Sections III–IV).
- An implementation study showing that moving the vector register file and scratchpad into distributed RAM reduces LUT usage by 5.7 $\times$ and closes 100 MHz timing, with a clear analysis of why the register-file read multiplexer, not the ALUs, dominates a naive SIMT datapath (Section IV).
- A six-kernel benchmark suite, verified bit-exact at up to 1024 threads, that isolates distinct microarchitectural behaviors, and a *measured* speedup against the scalar host—including a deliberately reported case where SIMT loses (Section VI).
- A divergence-aware lane clock-gating energy model that reuses the SIMT active mask as a per-lane clock-enable and quantifies the lane-datapath energy saved as a function of control divergence (Sections III, VI).

II. BACKGROUND: THE SIMT EXECUTION MODEL

A SIMT machine groups scalar *threads* into *warps* that share one instruction stream and execute on a vector of *lanes*. SIMTiX uses a warp size equal to its lane count ($\text{WARP_SIZE} = \text{NUM_LANES} = 8$), so one warp occupies all lanes for one instruction. Multiple warps are resident simultaneously ($\text{NUM_WARPS} = 4$) and time-share a single fetch/decode/execute datapath; when one warp stalls on memory, the scheduler issues another, hiding latency. Three effects determine SIMT efficiency: *occupancy* (enough warps to hide latency), *memory coalescing* (merging lanes' accesses into wide transactions), and *control divergence* (lanes within a warp taking different branch outcomes, which serializes the divergent paths). SIMTiX implements hardware for all three and instruments each so they can be measured.

III. SIMTiX ARCHITECTURE

SIMTiX is an offload accelerator. The host RISC-V core programs a command block over a memory-mapped (MMIO)

register interface—kernel entry PC, base pointers A/B/C, and thread count N—then writes a GO bit and polls a DONE status bit. On GO the accelerator launches a grid of N threads as $\lceil N/8 \rceil$ warps. Each thread receives its identity and arguments through a register-seeding convention at spawn (a0=thread id, a1 . . a3=base pointers, a4=N).

A. Warp pool, scheduler, and latency hiding

The warp pool holds four resident warp slots, each with its own architectural state (program counter, active mask, register file partition, and reconvergence stack), all sharing one datapath and one data port. A round-robin scheduler issues one runnable warp per cycle. Memory instructions are handed to a *background memory engine* that services the access while the scheduler keeps issuing *other* warps’ compute instructions on the otherwise-idle datapath—the classic GPU latency-hiding trick. At most one memory instruction is in flight (single data port); a warp that wants memory while the port is busy simply waits its turn without blocking others.

B. Per-lane datapath and the register file

Each lane is a single-cycle RV32IM integer datapath (add/sub, logic, shifts, set-less-than, and mul; no divide or floating point). The vector register file (VRF) is logically $\text{NUM_WARPS} \times \text{NUM_LANES} \times 32$ words. A naive flip-flop VRF is the dominant cost of the design (Section IV); SIMTiX instead implements it as one distributed-RAM bank per lane, addressed by {warp, register}, with one synchronous write port and two asynchronous read ports.

C. Memory coalescing

The data port is one cache line wide ($\text{LINE_WORDS} = 8$, 256 bits). When a warp issues a load or store, the memory engine groups the active lanes’ effective addresses by line and spends one transaction per distinct line. A contiguous warp access (e.g. $A[\text{tid}]$) collapses to a single line transaction, while a fully scattered access costs up to eight. A debug counter records line transactions, making coalescing efficiency directly measurable.

D. Control divergence: the reconvergence stack

Each warp owns a small SIMT reconvergence stack. The top-of-stack frame defines the warp’s live PC and *active mask* (which lanes execute). A branch is resolved per lane: if all active lanes agree, the PC is simply retargeted; if they disagree, a frame is pushed so the two lane groups run in turn and reconverge at a join PC. SIMTiX supports the compiler-free convention of single-sided *if* and divergent loops: a forward branch reconverges at its target, a backward branch at the fall-through. A divergence counter records each divergent branch event.

E. On-chip scratchpad

A per-warp scratchpad (64 words/warp) occupies a dedicated address aperture. Accesses that fall in the aperture are serviced from on-chip RAM and never reach the global data port, so data reused across a warp’s lanes (for example a

TABLE I
ACCELERATOR PPA ON XILINX XCZU7EV BEFORE/AFTER MOVING THE VRF AND SCRATCHPAD TO DISTRIBUTED RAM (OUT-OF-CONTEXT SYNTHESIS).

Metric	FF-based VRF	LUTRAM VRF+scratch	
	baseline	after	Δ
LUTs	170,868 (74%)	30,088 (13%)	−82%
Flip-flops	44,112 (10%)	4,590 (1.0%)	−90%
DSP48E2	24	24	—
BRAM	0	0	—
Timing (100 MHz)	WNS −3.21 ns	WNS +2.25 ns	met
Power	1.67 W @200 MHz	0.81 W @100 MHz	—

broadcast matrix row) can be staged once and reused, cutting global traffic.

F. Divergence-aware lane clock-gating

The active mask is exactly the set of lanes doing useful work; masked-off lanes produce no architectural change yet, in a baseline design, still clock their pipeline registers. SIMTiX treats the active mask as a per-lane clock-enable: the mask drives an integrated clock-gate on each lane’s registers, so only active lanes are clocked. Let an issued instruction activate a of L lanes. An ungated design clocks L lanes; a gated design clocks a . Summing over a kernel, the lane-datapath dynamic energy a gated design saves is

$$E_{\text{saved}} = 1 - \frac{\sum_i a_i}{L \cdot \sum_i 1} = 1 - \frac{\text{active-lane instructions}}{L \cdot \text{issued instructions}}. \quad (1)$$

Two hardware counters—issued instructions and summed active lanes—make Eq. (1) directly observable.

IV. IMPLEMENTATION

SIMTiX is written in synthesizable SystemVerilog and integrated with a reused five-stage RV32I host pipeline into a self-contained chip (`chip_top`): host core, driver ROM, accelerator, and an eight-bank on-chip shared memory. In the integrated chip the host loads the operands into shared memory, launches the accelerator, reads the results back, and publishes a checksum—the whole CPU→memory→accelerator→memory→CPU loop runs in hardware.

A. From flip-flops to LUTRAM

The first synthesis of the accelerator on a Zynq UltraScale+ (xczu7ev) consumed 170,868 LUTs—74% of the device—at only 10% of its flip-flops, and topped out near 122 MHz. The cause is structural: a flip-flop VRF of $4 \times 8 \times 32$ 32-bit words is 32,768 registers read as `vrf[warp][lane][reg]` with *both* warp and register variable, i.e. a 128-to-1 multiplexer per lane per read port. That multiplexer fabric, not the ALUs, dominates area and sets the critical path. Re-expressing the VRF as one distributed-RAM bank per lane, and likewise moving the scratchpad into LUTRAM, removes essentially all of those registers. Table I summarizes the effect: LUTs fall by $5.7\times$ and the design closes 100 MHz timing with positive slack.

TABLE II
BENCHMARK KERNELS AND THE MECHANISM EACH ONE STRESSES.

Kernel	Computation	Stresses
vadd	$C_i = A_i + B_i$	baseline streaming
saxpy	$C_i = 3A_i + B_i$	coalescing, mul
fir	$C_i = A_i + 2A_{i+1} + A_{i+2}$	stencil, partial coalescing
relu	$C_i = \max(0, A_i)$	data-dependent light divergence
collatz	Collatz step count	data-dependent heavy divergence
reduce	$C_w = \sum_{l=0}^7 A_{8w+l}$	scratchpad + reduction

B. Full-chip place-and-route

The complete chip was taken through opt/place/phys-opt/route to a bitstream on xc7ev. Post-route it meets all timing at 100 MHz (setup WNS +0.000 ns, hold met) and uses 34,850 LUTs (15.1%), 6,130 flip-flops (1.3%), 3,348 LUTRAM, 24 DSP, and zero BRAM, at 1.09 W. The accelerator is $\sim 87\%$ of the logic; the host core and 4 KB shared memory are small by comparison.

C. Verification ladder

Correctness is established at four levels: (i) cycle-accurate RTL simulation with self-checking testbenches (Verilator); (ii) behavioral simulation in the vendor GUI; (iii) full place-and-route with post-route timing sign-off; and (iv) post-implementation *gate-level* functional simulation of the routed netlist (UNISIM primitives), which reproduces the integrated chip’s golden checksum of 964 bit-for-bit. The benchmark suite below adds a fifth, breadth-oriented level.

V. METHODOLOGY

A. Benchmark suite

We wrote six kernels, each chosen to stress a distinct mechanism (Table II). All are assembled from RISC-V source with a stock toolchain. Each runs at $N \in \{8, 64, 256, 1024\}$ threads (the divergence-heavy *collatz* to 256), and every output element is checked against a golden model computed in the testbench.

B. Metrics

Per run the engine reports cycles (launch to done), global line transactions, scratchpad transactions, divergent-branch events, issued instructions, and summed active lanes. From these we derive lane utilization $\sum a_i / (L \sum 1)$ and a scalar-equivalent throughput $\sum a_i / \text{cycles}$ (work retired per cycle).

C. Measured scalar baseline

For an apples-to-apples speedup, each kernel is also implemented as a natural scalar loop and executed on the *same* five-stage host pipeline in RTL, counting cycles from program start to completion. The host is pure RV32I, so constant multiplies are lowered to shift-and-add; *matmul*, which needs a general multiply, is therefore excluded from the scalar comparison. The scalar program and the accelerator are checked to produce identical results.

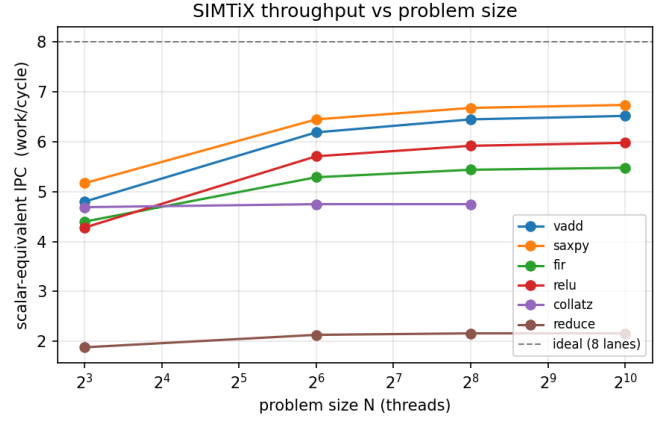


Fig. 1. Scalar-equivalent throughput versus problem size. Throughput ramps as occupancy rises (latency hiding) and saturates below the 8-lane ideal due to the single shared data port.

VI. RESULTS

A. Throughput and latency hiding

Figure 1 plots scalar-equivalent throughput against problem size. With only eight threads (one warp) throughput is low—there is nothing to hide memory latency behind—but it climbs as more warps become resident and saturates near 6.5–6.7 work-items/cycle for the coalesced kernels, out of an ideal of 8. The gap is the single data port: while one warp’s coalesced access occupies the memory engine, only three other warps are available to fill the datapath.

B. Memory coalescing

Figure 2 shows global line transactions versus size. *saxpy/vadd* are perfectly coalesced (three line transactions per warp—one each for A, B, C), whereas the *fir* stencil reads $A[i], A[i+1], A[i+2]$; the shifted reads straddle line boundaries and double its traffic. This is the expected signature of *partial* coalescing and explains *fir*’s lower speedup. A deliberately scattered access (stride 8) costs the full eight transactions per warp—an $8\times$ traffic increase that coalescing removes for contiguous data.

C. Control divergence and lane utilization

Figure 3 reports lane utilization. The convergent streaming kernels keep all eight lanes active every instruction (100%). *relu* diverges only where the input changes sign, dropping utilization slightly to 93.7%. *collatz*, whose per-lane trip count is data-dependent, falls to 68.3% as finished lanes mask off while others continue. *reduce* is lowest (39.8%) because its second phase collapses eight lanes’ partial sums on a single lane. Per Eq. (1), the unused fraction is the lane-datapath dynamic energy a clock-gated design saves: from 0% on convergent code up to 22.5% on a divergence-heavy microbenchmark.

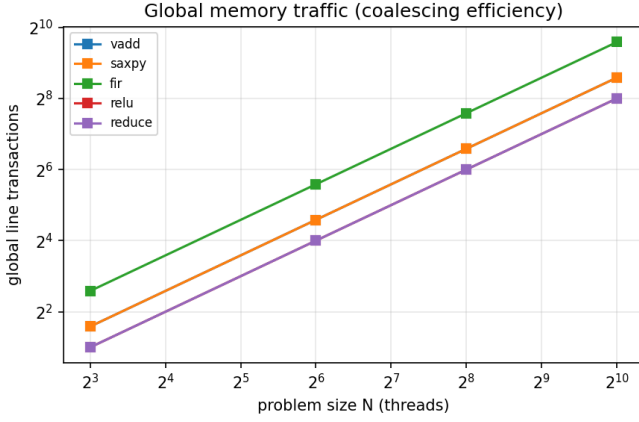


Fig. 2. Global line transactions versus size. Contiguous kernels coalesce to three transactions per warp; the `fir` stencil’s shifted reads straddle lines and roughly double the traffic.

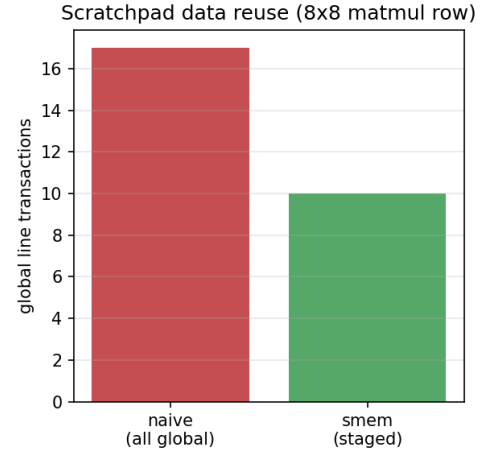


Fig. 4. Global line transactions for an 8×8 matmul row: staging the reused row in the scratchpad cuts off-chip traffic.

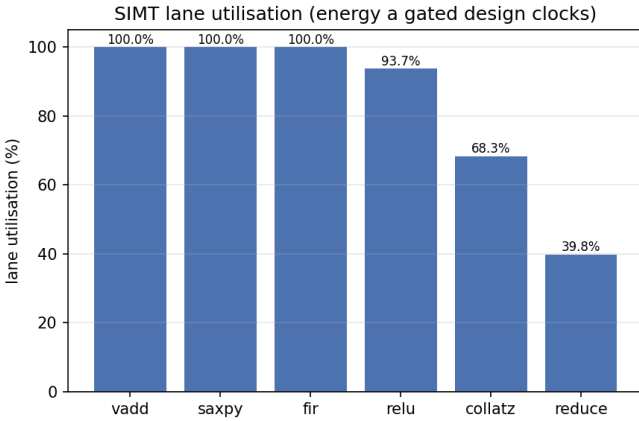


Fig. 3. SIMT lane utilization per kernel (largest size). The complement is the lane-datapath dynamic energy a divergence-aware clock-gated design saves (Eq. (1)).

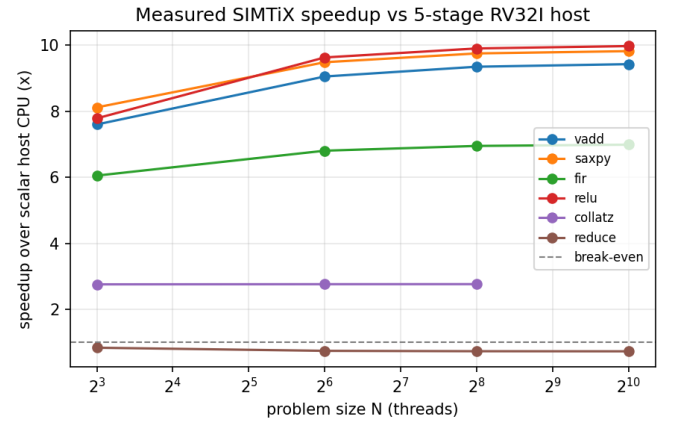


Fig. 5. Measured speedup over the five-stage RV32I host running the same work. Streaming kernels reach 7–10 $\times$; divergence reduces the gain; the serial reduction falls below break-even.

D. Scratchpad data reuse

For an 8×8 matrix-row product, staging the broadcast input row in the scratchpad once and reusing it cuts global line transactions from 17 to 10 (Figure 4). At this small size the staged version spends more cycles (the scratchpad is single-ported and serializes the stage-in), so the win is in off-chip traffic rather than latency—an honest trade that grows with data size and reuse distance.

E. Measured speedup over the scalar host

Figure 5 and Table III give the headline result: measured accelerator speedup over the scalar host executing identical work. Streaming kernels reach 7–10 $\times$, approaching the eight-lane width scaled by the coalescing traffic reduction. `fir` is lower (7.0 $\times$) because of its partial coalescing. `collatz` achieves 2.8 $\times$: heavy data-dependent divergence erodes—but does not eliminate—the SIMT advantage. `reduce` is below unity (0.72 $\times$): its single-lane collapse and single-ported

scratchpad make it *slower* than the scalar core. We report this case deliberately—the SIMT model is not universally beneficial, and a reduction needs a better hardware primitive (e.g. a tree-reduction network) to win on this engine.

F. Design-space exploration: lanes and warps

SIMTiX is parameterized in its lane count and resident-warp count (`WARP_SIZE = NUM_LANES`, and the coalescing line scales with the lanes). We swept `NUM_LANES` $\in \{4, 8, 16, 32\}$ against `NUM_WARPS` $\in \{2, 4, 8\}$ —all twelve configurations elaborate and pass the suite bit-exact—and measured throughput on the five lane-count-agnostic kernels. Table IV reports the geometric-mean throughput, and Figure 6 plots it against lane count.

Two clear effects emerge. First, throughput scales almost linearly with lanes: $3.06 \rightarrow 5.85 \rightarrow 11.68 \rightarrow 23.24$ work-items/cycle, about $1.97\times$ per doubling, i.e. $7.6\times$ from 4 to 32 lanes. Widening the lanes also widens the coalescing line, so

TABLE III
MEASURED CYCLES AND SPEEDUP (LARGEST SIZE; `COLLATZ` AT $N=256$). LANE UTILIZATION AND GLOBAL TRAFFIC FROM THE ACCELERATOR COUNTERS.

Kernel	N	Accel	CPU	Speedup	Lane-util
<code>vadd</code>	1024	1,413	13,322	$9.43\times$	100%
<code>saxpy</code>	1024	1,669	16,394	$9.82\times$	100%
<code>fir</code>	1024	2,053	14,345	$6.99\times$	100%
<code>relu</code>	1024	1,284	12,809	$9.98\times$	93.7%
<code>collatz</code>	256	9,637	26,601	$2.76\times$	68.3%
<code>reduce</code>	1024	6,023	4,363	$0.72\times$	39.8%

TABLE IV
GEOMETRIC-MEAN THROUGHPUT (WORK-ITEMS/CYCLE) OVER THE FIVE LANE-AGNOSTIC KERNELS, SWEEPED OVER LANES AND RESIDENT WARPS. ALL TWELVE CONFIGURATIONS VERIFY BIT-EXACT.

NUM_LANES	NUM_WARPS		
	2	4	8
4	3.06	3.06	3.05
8	5.85	5.85	5.85
16	11.68	11.68	11.65
32	23.24	23.24	23.14

global line transactions fall proportionally (e.g. `vadd` drops $768 \rightarrow 96$ transactions from 4 to 32 lanes). Second, and more interesting, the resident-warp count makes essentially no difference here (the three curves in Figure 6 coincide; eight warps is marginally *worse* due to extra spawn bookkeeping). This is honest about the harness: the behavioral memory model answers in a single cycle, so there is almost no latency for extra warps to hide; occupancy beyond the two warps needed to overlap the memory engine with the datapath pays off only under realistic multi-cycle memory. The two knobs are thus separable—lanes buy data-parallel throughput, warps buy latency tolerance—and both cost register-file area ($\propto \text{lanes} \times \text{warps}$), which motivates the area/frequency Pareto study (Section VIII).

VII. RELATED WORK

Open RISC-V GPGPU cores include Vortex [1], a full OpenCL-capable multi-core design, and Ventus [2], a PTX/Vortex-style GPGPU; both target performance and scale and are correspondingly large. Simty [3] demonstrates generalized SIMT execution of the RISC-V ISA, and Nyuzi [4] is a synthesizable vector GPGPU for graphics research. Relative to these, SIMTiX is deliberately small and readable, and its contribution is not raw performance but a complete, reproducible RTL-to-bitstream flow paired with a measured, mechanism-isolating evaluation and an energy-aware divergence study. Control-divergence handling via reconvergence stacks and dynamic warp formation is long studied [5], [6]; we use the standard stack mechanism and instead *measure* its energy implication through mask-driven lane clock-gating.

VIII. DISCUSSION AND FUTURE WORK

The evaluation points to concrete next steps. The single shared data port caps throughput below the lane-count ideal;

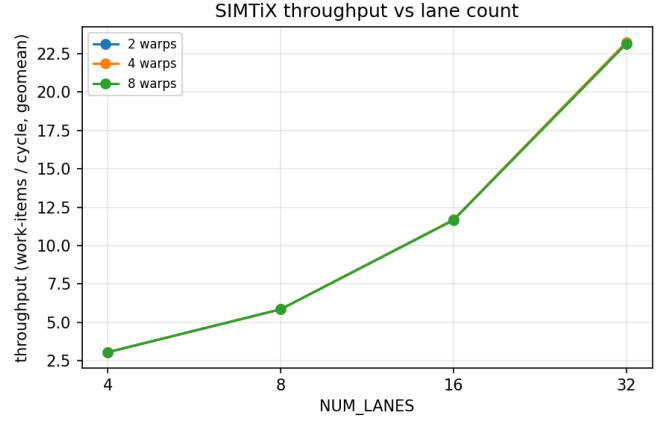


Fig. 6. Throughput versus lane count for each resident-warp count. Throughput scales near-linearly with lanes; the warp-count curves coincide because the single-cycle memory model offers little latency to hide.

multiple memory ports or banked access would raise the saturation point. The `reduce` result motivates a dedicated cross-lane reduction primitive. The lane clock-gating model is analytic from counters; a gate-level power simulation would confirm the realized savings. The throughput design-space sweep (Section VI-F) establishes how performance scales with lanes and warps; the complementary piece in progress is a per-configuration FPGA and ASIC PPA sweep to complete the area/frequency/throughput Pareto. Finally, an open-source ASIC path through the SkyWater 130 nm PDK [7] using OpenLane [8] would turn the FPGA PPA numbers into silicon-relevant area, frequency, and power.

IX. CONCLUSION

SIMTiX shows that the full SIMT mechanism set—warp scheduling and latency hiding, memory coalescing, a reconvergence stack, and a scratchpad—fits in a compact, fully-open RISC-V accelerator that runs end-to-end through a real implementation flow. A distributed-RAM register file is the key to making it small and fast on FPGA. Across a verified six-kernel suite, SIMTiX delivers 7–10 $\times$ measured speedup where the SIMT model fits, degrades gracefully under divergence, and—reported honestly—loses on a serial reduction, while its mask-driven clock-gating saves up to 22.5% of lane-datapath dynamic energy on divergent code. The design and its evaluation infrastructure are public, so the results are reproducible and the platform is reusable for teaching and research.

REFERENCES

- [1] B. Tine, K. P. Yalamarthy, F. Elsabbagh, and H. Kim, “Vortex: Extending the RISC-V ISA for GPGPU and 3D-Graphics,” in *Proc. IEEE/ACM MICRO*, 2021.
- [2] F. Zhang *et al.*, “Ventus: a RISC-V based GPGPU,” open-source project, 2022.
- [3] S. Collange, “Simty: Generalized SIMT execution on RISC-V,” in *Workshop on Computer Architecture Research with RISC-V (CARRV)*, 2017.
- [4] J. Bush, P. Dexter, T. N. Miller, and A. Carpenter, “Nyami: a synthesizable GPU architectural model for general-purpose and graphics-specific workloads,” in *Proc. IEEE ISPASS*, 2015.

- [5] W. W. L. Fung, I. Sham, G. Yuan, and T. M. Aamodt, "Dynamic Warp Formation and Scheduling for Efficient GPU Control Flow," in *Proc. IEEE/ACM MICRO*, 2007.
- [6] E. Lindholm, J. Nickolls, S. Oberman, and J. Montrym, "NVIDIA Tesla: A Unified Graphics and Computing Architecture," *IEEE Micro*, vol. 28, no. 2, 2008.
- [7] SkyWater Technology and Google, "SkyWater Open Source PDK (sky130)," <https://github.com/google/skywater-pdk>.
- [8] A. Ghazy and M. Shalan, "OpenLane: The Open-Source Digital ASIC Implementation Flow," in *Workshop on Open-Source EDA Technology (WOSET)*, 2020.
- [9] A. Waterman and K. Asanović, "The RISC-V Instruction Set Manual, Volume I: Unprivileged ISA," RISC-V International.